

REPORT ON

**EVALUATION OF DYNAMIC PERFORMANCE OF A
ROBOTIC MANIPULATOR USING ARDUINO AND PWM
DRIVEN SERVO ACTUATORS**

Submitted in partial fulfilment of the requirement for the award of the degree of

**BACHELOR OF TECHNOLOGY
IN
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**DEPARTMENT OF
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CERTIFICATE OF COMPLETION

This is to certify that Challa Chaithanya Reddy (24095A0401), Pinjari Haroon Rasdi (23091A0456) of III B.Tech., I-Semester, have successfully completed the assigned Project work on “**Evaluation of Dynamic Performance of a Robotic Manipulator Using Arduino and PWM Driven Servo Actuators**” under the guidance of **Dr.D.Usen**. This Project was completed as part of the requirements for the course Electronics and Communication Engineering, during the academic session of III B.Tech., I-Semester.

We acknowledge the student's sincere efforts, timely submission and dedication towards the successful completion of this work.

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1. Abstract

ROBOTIC MANIPULATOR:

One of the most transformative technologies in the contemporary world has turned out to be robotics, which has taken the industrial progress in the sphere of automation, healthcare, defense, and education. Robotic manipulators have especially been effective in substituting monotonous and manual duties by precision, reliability, and effectual mechanized tasks. The commercially available robotic arms are however costly, complicated to use and demand sophisticated software or particular training thus not affordable to students, and small-scale developers.

To deal with these problems, this project examines how a simple robotic arm can be designed and built with minimal cost using Arduino UNO and servo motor and fully operated using simple text commands using serial..

The system is also characterized by the four servo motors, two of which are involved in the gripper movement (mirrored in order to operate in sync), one is the shoulder joint motor and the other the elbow joint motor. This design is opposed to joystick-controlled, or pre-programmed robotic arms by the addition of an adaptable text-based interface, wherein the user programs the Arduino Serial Monitor with a command to be read and executed by a robotic arm. All the commands are met with single servo response (ROR example: SHOULDER 120 or GRIP CLOSE) or an integrating, macro action (ROR example: RISE CLOSE or DOWN OPEN). The same macro commands are used to manage more than two servos in order to enter complex movements and provide greater usability and efficiency in terms of performance..

The project does not only offer students and hobbyists a cost-effective robotic arm solution but also creates a framework which can be enhanced with other functionalities like wireless communication, sensor integration or AI-powered vision system. Its learning potential is incalculable as it hinges the education with the practical implementation in embedded systems, control engineering, and robotics. The suggested robotic arm shows that highly developed robotic capabilities can be repeated at a lower cost, and affordable level allowing future engineers to develop and investigate the automation technologies without any financial obstacles.

2.Introduction

Robotics is an interdisciplinary science, encompassing mechanical engineering, electronics and computer science, to develop intelligent machines with the capability to supplement or substitute human labor in a variety of applications. Having started with heavy-duty industrial manipulators up to collaborative service robots, the role played by robotic systems in the contemporary world cannot be overemphasized. Most robotic systems revolve around the central component, that is the manipulator itself, an articulated arm that is geared towards interacting with the environment through the use of movement, lifting, or manipulating of objects in a precise, repeatable manner. Regarding the application of robotic manipulators, the challenges posed by high cost and complexity prevent their usage at the academic and educational level, yet their widespread application has transformed industries, including manufacturing, logistics, healthcare, and defense. Consequently, scientists and educators have resorted to low-end microcontrollers like the Arduino to develop robot prototypes, which are cost-effective and useful and enable students to have practical experience on robotics in real life. One type of robot arm is commonly actuated using servo motors, especially because they have the capability to perform precise angular positioning by means of Pulse Width Modulation (PWM) signals. Hence this project is comprised of 2 primary parts :

1.SERVO MOTOR(actuator)

2.ARDUINO UNO(controller)

1.SERVO MOTOR:



Fig 2.1:working of servo motor

A servo motor is a self-contained electrical machine which moves or moves components of a machine with a high level of efficiency and very precise. It is common

in robots, automation and in industry machinery where a fine degree of control is essential.

Working Principle:

The principle behind the actions of a servo motor is its accuracy in the control of angular position, velocity, and acceleration. It does this by the closed loop control system:

- **Motor:** This normally refers to a DC/AC motor (usually a brushless DC motor) that renders the torque needed to revolve the output shaft.
- **Gear Reduction Unit:** The vast majority of servos have a gear train which greatly decreases the amount of output speed and acts to augment the torque. This is why a small motor is able to work on a relatively heavy load.
- **Position Sensor (Encoder or Potentiometer):** This is an essential part that will be used in measuring the actual position of the motor shaft. For this a potentiometer i.e variable resistor is used in smaller and hobby servos. A high-resolution encoder is employed in industrial quality servos.
- **Control Circuit (Controller/Amplifier):** This is the brain circuit. It is driven by an external source (such as a microcontroller) with a desired position signal (the Command Signal) compared to that which the sensor is telling users its actual position is.
- **Error Signal and Feedback:** The control circuit calculates The error i.e $\text{Error in Servo position} = \text{Command Signal} - \text{Actual Position}$ is calculated by the control circuit. In case of any error (the motor is not in the correct spot), the controller produces the relevant correction signal to the motor to move it in the required direction. This mechanism is the closed-loop feedback system.

Controlling Servo Using PWM Signals:

Servos (such as in a RC hobby) with small common windings are usually driven by a signal commuted by a Pulse Width Modulation (PWM) signal:

- **PWM Signal:** The external controller sends a continuous stream of electrical pulses, usually at a fixed frequency (e.g., 50 Hz, meaning a pulse every 20 milliseconds).
- **Pulse Width:** The width of the positive part of this pulse dictates the desired angle.
 - A thin pulse (e.g., 1.0ms) may be used to move the motor to the 0 degree position.

- The motor may have a mid-range pulse (e.g. 1.5ms) to 90 degree (center) position..
- A broad pulse (e.g. 2.0 ms) could pull the motor to the -180 degree position.
- **Operation:** The control circuit on the inside of the servo motor accesses the width of an incoming pulse into a target position. The circuit then powers the motor until the sensor detects that the shaft is at that position and then continually makes corrections to ensure that shaft maintains that angle regardless of any external force.

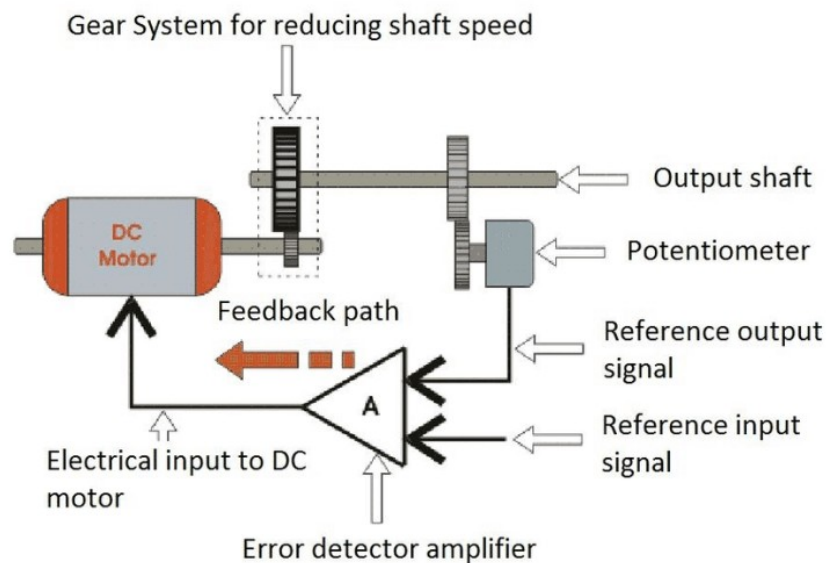
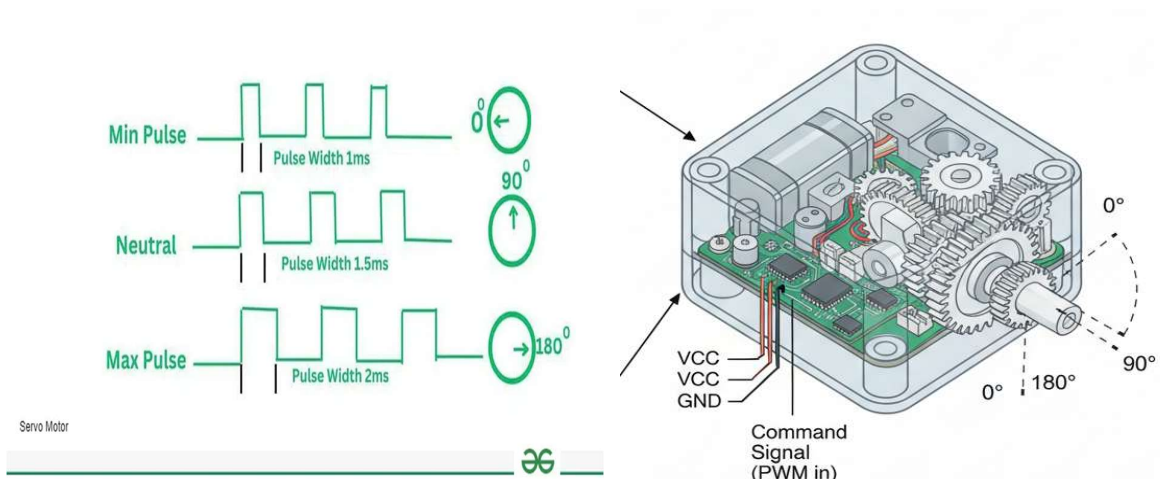


FIG 2.2 :Control of Servo motor

2.ARDUINO UNO

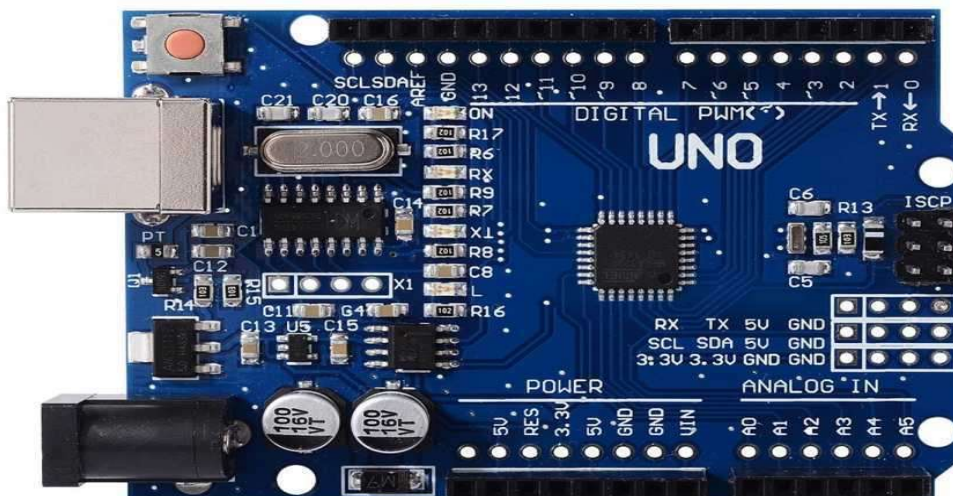


Fig 2.3 :Arduino uno

With these two elements, it is possible to create a robotic manipulator that is affordable, simple, and functional. The given project addresses these limitations by adding a text based serial command interface. The user is able to type commands . through the Arduino Serial Monitor. The contributions of this project can be summed up as follows:

- Design of a 3-DOF prototype of robotic arm,based on the Arduino UNO and servo motors.
- Application of text control of serial command control to provide accurate and versatile manipulation.
- Co-ordination of multi-servo functions..
- Blending of gradual progressive movement to attain natural and steady robotic action.
- Tests of the system through Wokwi simulation that will be done to verify the reliability of the system before deployment.

Seizing the gap between cheap educational robots prototypes and the advanced robotic manipulators, this project proves that robots can be inexpensive and at the same time, precision, flexibility, and usability may be attained. It can be used as a starting point of further improvements, including wireless control, feedback sensors or computer vision enabled autonomous operation. In the end, this paper discusses the need to have robotics education available to everyone and demonstrates how embedded systems can give students and hobbyists the power to be creative in the automation industry.



3.Literature Survey

Decades of research have added value to the field of robotics, and robotic manipulators have been the core of automation in both industrial and research fields. Robotic arm: This is a form of a mechanical manipulator that resembles a human arm in that it can pick, place, pick, or rotate an object.

The emergence of the low-cost robotic arms has caused a lot of interest in the academic community and mostly on the undergraduate level, as they offer practical experience with regard to the control systems, programming, and embedded electronics.

In this section, a review of the literature will be provided and how the proposed robotic gripper arm, which is based on arduino, has solved the gaps. Early work on the use of robotic manipulators focused on giant industrial power tools that could be operated by means of high-powered stepper or DC motors. These systems were made according to the heavy duty operations like automotive assembly line, welding, and handling of materials. Although they were efficient, they were very expensive and complicated and as such, could not be used in education or experimental research.

However, with time researchers started experimenting with smaller scale models, with hobby sized parts such as servo motors which are small, cheap, and can be made to rotate with high angular resolution with PWM signals. Multiple sources of research have demonstrated and explored the use of joystick-controlled robotic arms in the laboratory and classroom environment due to affordability and functionality.

The analog joysticks in these models are said to give feedback that is proportionate to the angles of the servo motors. This system is the same technique as putting in some intuition and works manually on each joint. To illustrate, a robotic arm controlled by a joystick was shown performing object manipulation tasks in one of the projects given at IEEE Student Conference on Robotics. Though, the greatest drawback of the joystick-based control is the inability to be precise. Joystick movements are prone to human errors hence poor repeatability. There is a significant weakness because operators cannot always cause the arm to go to the same position, which is vital when it has to be precise. Moreover, complicated actions demand simultaneous manipulation of different joysticks, which are overloading the user with cognitive tasks. By contrast, pre-programmed robot arms have gained broad usage as well. With this type of model, the programmer predetermines a sequence of servo positions.

4.Problem Statement

Performance assessment of an Arduino and PWM-controlled Servo Actuator-controlled Robotic Manipulator:

The modern automation systems have become impossible without the use of robotic arms and they have transformed industries by making them efficient in repetitive, hazardous, and precision demanding jobs. Nevertheless, with educational or low-cost prototypes, a number of challenges still exist which reduce their accessibility, usability, and performance. This section precisely presents issues in this project and explains the objectives that will be followed in its design and implementation..

Challenges of project:

1. The initial one is the lack of accuracy in control mechanisms. Various low-cost robotic arms prototypes are based on joystick-type interfaces, in which analog joystick inputs are linearly transformed to servo positions. This is intuitively correct, but it brings according inconsistencies. Small stick movements can cause large changes in an angle of a servo and therefore, it is virtually the same, positioning the stick to repeatable and accurate movements. Such inconsistencies in real-world applications compromise the use of the arm, e.g. picking up small objects or aligning components.
2. .The second issue is low flexibility of pre-programmed systems. The other method commonly used in the academic robotic arms is to program a sequence of servo angles in the Arduino code, which the arm will follow when switched on. These systems are inflexible though precise since to modify a movement, one has to write new code and re-upload it again. This inflexibility does not allow users to dynamically experiment and this is essential in the learning context and in prototyping.
3. .The third problem is connected with the lack of the macro-level coordinated control of most of the student level robotic arms. High-level commands like pick or place are generally used by industrial manipulators and they coordinate several joints at once. Nevertheless, issues of control in educational prototypes are frequently reduced to a single servo at most. The limitation complicates the usage of the systems where only a user can control multiple joints after which a basic task should be accomplished. To take an object as an example, the user would need to command



the shoulder to lift separately, the elbow to be moved, and the gripper to be closed and that too consecutively. As far as the macro-level instructions are not provided, the usability is deteriorated, and the work of the operator who has to perform some routine has to be paid so much money and effort are wasted.

4. The fourth issue is some jerky or sudden motion due to direct assignment of servo angles. Two angles commanding a servo to move cause the servo to jump to the target at once and this causes abrupt jumps. This suddenness is able to impose mechanical loads on the servos, decrease precision, and render the arm incapable of working with delicate items. This is solved by industrial robots via trajectory planning and smooth motion control, which are not frequently available in low-cost educational designs since they are often complex..

Objectives of the Project:

This project aims at designing and implementing a Arduino UNO and servo motor based 3-DOF robot arm controlled through a serial command control.

The system aims to:

- Offer accurate and repressible control with a text based command.
- Support real-time flexibility, by admitting the user inputs, without reprogramming.
- Introduce macro commands for coordinated multi-joint movements i.e implementing several moves sequentially with one command.
- changing steps gradual servo movement to kind of give realistic robot movement.
- Test the system in Wokwi by simulation then test it in the field.

These issues and goal achievements enable the project to lead to the creation of a low-cost but high-tech robotic prototype, which may find its use in the sphere of education, experimental, and prototype applications.

5. Proposed Methodology

This project will use a robotic arm, three degrees of freedom (DOF), as the proposed structure designed and conducted using an Arduino UNO microcontroller and servo actuators. The main idea is to combine a serial command interface enabling the user to issue straightforward text-based commands to the arm to move in either joint-by-joint mode or as a result of a set of macro commands. The hardware implementation, software design, controlling algorithms, as well as simulation validation are all mentioned in the methodology..

Hardware Design :

- **Servo 1 (BASE)** :regulates the ROTATORY movement of the arm, and it looks like the movement of the arm after turning right or left.
- **Servo 2 (SHOULDER Joint):** Also controls the vertical list motion of the arm like the arm is lifted/ lowered by the shoulder.
- **Servo 3 (Gripper):** It involves one servo that can be used to open or close the grip with the help of a fixed limb..

Power is supplied to the servos on PWM pin of the Arduino, external power supply may be supported as there are several servos, which consume a large current. The control pins of choice are D3, D5, D6 and in that order,base,shoulder, gripper respectively. The Arduino UNO serves as the primary controller that serves to receive the input commands through the Serial Monitor via USB device. The commands are combusted and converted to the PWM signals to move servos.

Software Design :

It is written in the Arduino IDE having the standard library of Servo.h to control the motors. The code has the following functional modules:

1. **Serial Command Parser:** Scans the contents of the Serial Monitor and is identifying a command is either acting single (for example: SHOULDER 120, ELBOW 90, GRIP CLOSE) or content of a macro (RISE CLOSE, DOWN OPEN).
2. **Single Control Functions:** Each of the joints have an option to be controlled separately by specifying precise angles. E.g. SHOULDER 100 positions the shoulder servo to 100. Similarly GRIP OPEN, GRIP CLOSE commands are used to switch between open and closed functions of gripper.

3. **Macro Command Functions:** Higher level Put up multiple servant movements using the high level commands. An example of this is the RISE CLOSE, which positions the shoulder and elbow in an upward direction with one smooth coordinated motion of closing the gripper. This lowers the workload of the operators and offers the simulation of tasks in the real world.
4. **Smooth Motion Algorithm:** The servo angles are moved with a series of very small steps to achieve a gradual increase, thereby the movement is natural and smooth rather than having jumps. As an example, when a servo rotates between 60 and 120 degrees the step will be 1 o C per step with a brief rest period, eliminating mechanical stress and allowing the action to look more realistic.
5. **Status Feedback:** We get feedback of angles of servos After executing a command, and the Arduino prints the updated angles of each joint to the Serial Monitor. This guarantees the transparency and enables the movements to be debugged.

Control Flow :

A summary of the process flow will be as below:

1. **Input:** Serial Monitor takes up command on entry by the user.
2. **Parsing:** Arduino interprets command type (individual vs macro).
3. **Execution:** This is through gradual movements of servo motors to aim angles.
4. **Feedback:** The final positions are shown at Serial Monitor.
5. **Loop:** This system waits two command actions.

Such open organization guarantees the responsiveness and flexibility in real time.

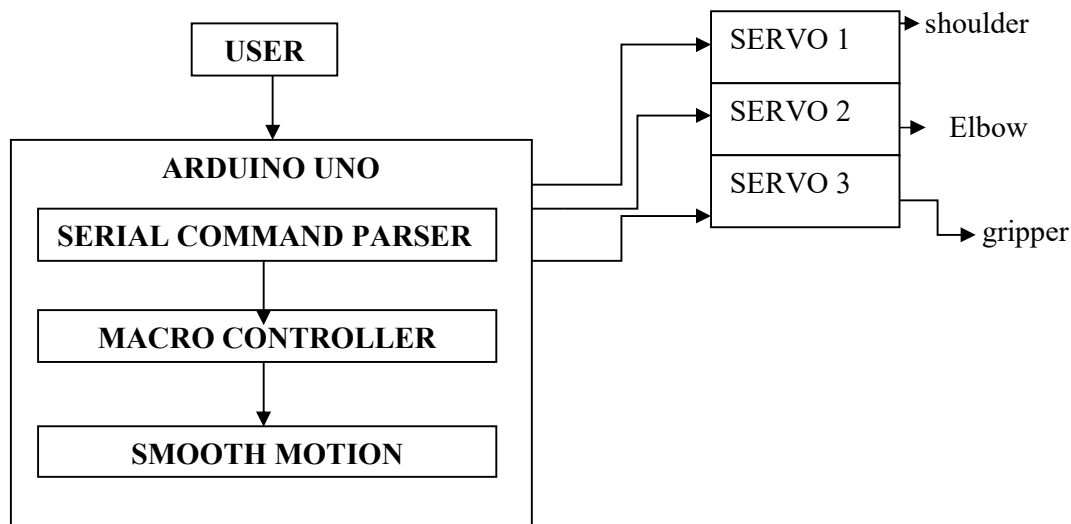


Fig 5.1:Block diagram of robotic maipulators

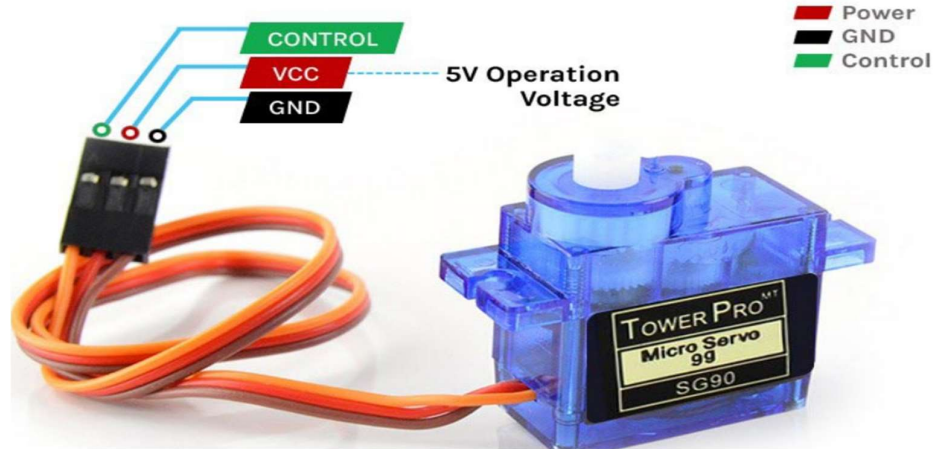


Fig 5.2: Pin interface of servo

Simulation with Wokwi :

The system is simulated with Wokwi, an online platform simulator of Arduino, before being prototyped physically. To check pin assignment, controlling PWM, and code logic, the Arduino UNO and four servos are virtually attached. Simulated tests are done with the simulated Serial Monitor and the behavior of the servo is visualized. This minimizes the chance of error in wiring, damages to the hardware and makes sure that the control algorithms operate properly.

The simulation in Wokwi provides the possibility of testing commands (both single and macro) in an iterative manner. Indicatively, the command MOVE 70 120 40 was able to move the shoulder to 70 o, the elbow to 120 o, and the gripper to 40 o successfully. On the same note, there were no errors of command on macro commands such as DOWN OPEN which would confirm the reliability of the algorithm.

Methodology Summary :

The suggested methodology is a needed combination of hardware, software, and simulation:

- **Hardware:** arduino UNO and 3 servos connected in a more suitable way.
- **Software:** A serial command parser and individual and macro functions.
- **Control:** Smooth incremental servo motion to create lifelike and stable movement.
- **Verification:** Simulation in Wokwi prior to physical implementation.

Such a methodology makes the robotic arm cheap, versatile, accurate, and powerful.

6.Implementation

The robotic arm project was implemented in three significant steps, which were hardware assembly, software development and simulation/testing. All the phases were well sequenced to be accurate and with good functionality as well as a strong system.

Hardware Implementation:

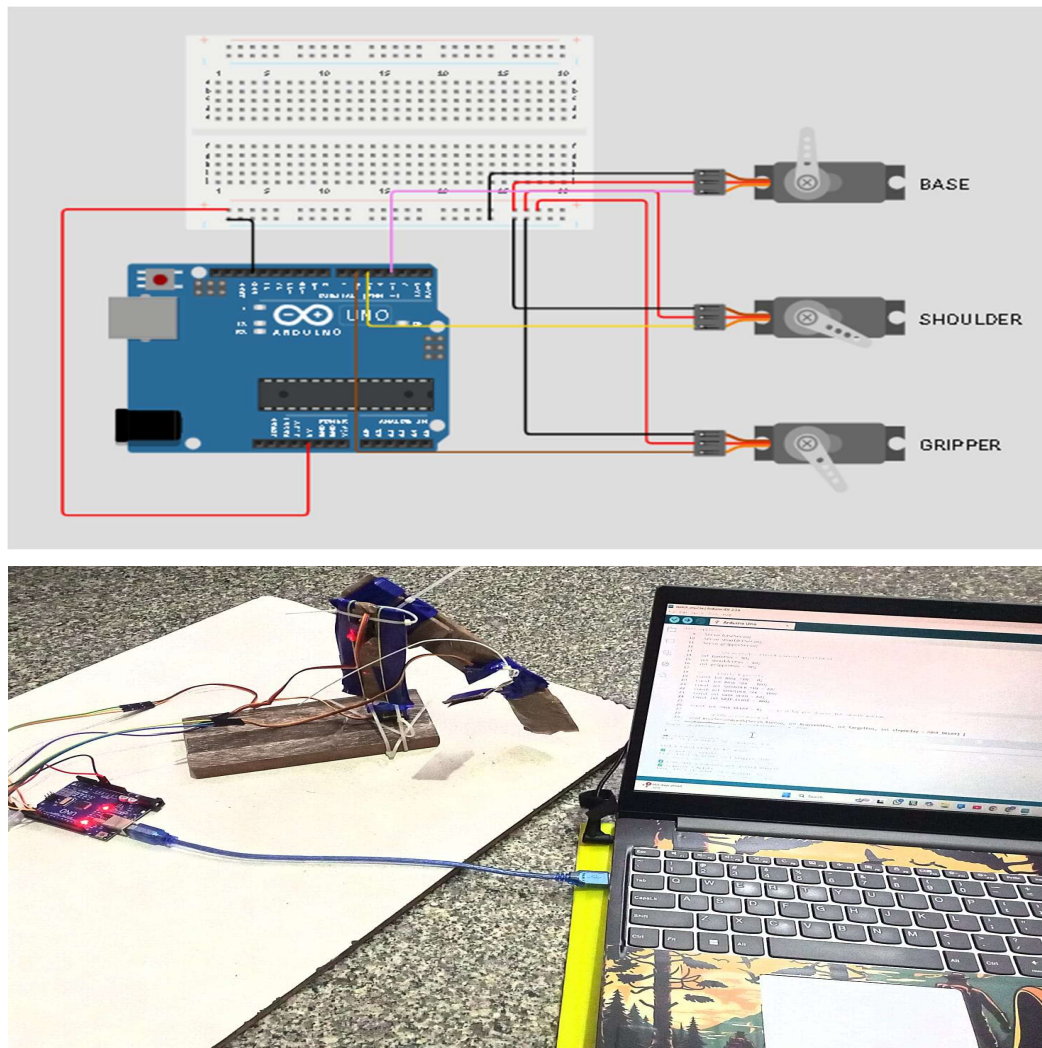


Fig 6.1: circuit diagram & Hardware Implimentation

The hardware design will have the four servo motors connected to the Arduino UNO board. The following is the set-up:

- **Servo 1 (Shoulder Joint)** → Digital Pin 3
- **Servo 2 (Elbow Joint)** → Digital Pin 5
- **Servo 3 (Gripper)** → Digital Pin 6

The servo consists of three wires VCC (red), GND (black/brown), and Signal (yellow/orange). All the servos VCC are connected to the 5 V line of the Arduino (or

another source of power in case of higher moment servos). The GNDs are coupled in a way that they are linked to the ground of the Arduino, which provides a common ground in all of them. The signal pins are wired into the corresponding digital pins which are capable of PWM.

The Arduino power regulator was not overloaded since several servos can consume a lot of currents when used simultaneously. Practically, the servos should be fed by an external 5 V controlled power supply of adequate current capacity (0.2A or more).

Software Implementation:

```
=== 3-Servo Robot Arm Ready ===
Commands:
  RIGHT <angle>    LEFT <angle>
  UP <angle>       DOWN <angle>
  OPEN            CLOSE
  RISE_CLOSE      DOWN_OPEN
  CENTER          STATUS
-----
You can chain commands like:
  RIGHT 45 RISE_CLOSE
  LEFT 30 DOWN_OPEN
-----
Base rotated RIGHT to 135
⚠ Unknown command: RISE
Gripper CLOSED
✅ Sequence complete.
Base: 135 | Shoulder: 90 | Gripper: 60
```

Fig 6.2 : Serial Monitor

The Arduino IDE has been used to write the control software through which the work of generating PWM signal is simplified by using the library of the name Servo.h. The code structure consisted of a number of modules:

1. **Initialization:** Each servo is connected to its pin, as well as are the initial positions (90 degrees neutral position).
2. **Command Input and Parsing** This is where the recipient of inputs(forms as commend) is the Serial monitor. The code removes empty areas, changes the text to capital letters and determines the type of command.
3. **Individual Servo Control:**

Command	Description
BASE 120	Move base servo to 120°
SHOULDER 60	Move shoulder servo to 60°
GRIPPER 150	Move gripper to 150°
CENTER	Move all to 90° center
STATUS	Show current positions



Base 120 or shoulder 60 level commands are directly tied to rapping the servo angles. GRIP OPEN and GRIP CLOSE indicate fixed angle values of grippers (e.g. the angle of grippers is taken as 40 o to open and 120 O to close).

4 **.Macro Commands** : Such commands as RISE CLOSE or RISE OPEN, DOWN CLOSE or DOWN OPEN result in predefined coordinated movements. Take the case of RISE CLOSE, the shoulder and elbow are elevated, but at the same time the grippers are closed.

RIGHT 45 RISE_CLOSE	Rotate base 45° right, then lift and close
LEFT 30 DOWN_OPEN	Rotate base 30° left, lower and open
UP 15 CLOSE	Raise shoulder 15° and close gripper

5. **Smooth Movement Algorithm** : A loop works and pushes or pulls the servo outputs in steps of 1° with short pauses (510ms) between them. This creates smooth, natural movement as opposed to start-and-stop motions, and the arm is made to be more believable.

6. **Feedback Output:** Once the implementation is completed the Arduino will provide real-time feedback on the servo positions after execution and send it to the serial monitor to verify and debug it

7.Results Analysis

Simulation Testing in Wokwi

The system was introduced in single-physically assembled Wokwi, a web-based simulator of the Arduino environment. Wokwi offers virtual Arduino boards and servo-motors so that testing of code can be done without a hardware.

The simulation verified:

- Initialization of servo correctly on start-up.
- Accuracy of movements for individual commands (SHOULDER 120, ELBOW 90, GRIP OPEN).
- Execution of macro-commands (RISE CLOSE, DOWN OPEN).
- The control of smooth movements between servos instead of being jerky.
- Status messages displayed at Serial Monitor were correctly matching servo described angles.

This step of simulation was the most important since the mistakes were avoided in advance of the hardware deployment. It as well approved the realisation of macro-level control and tender movement logic.

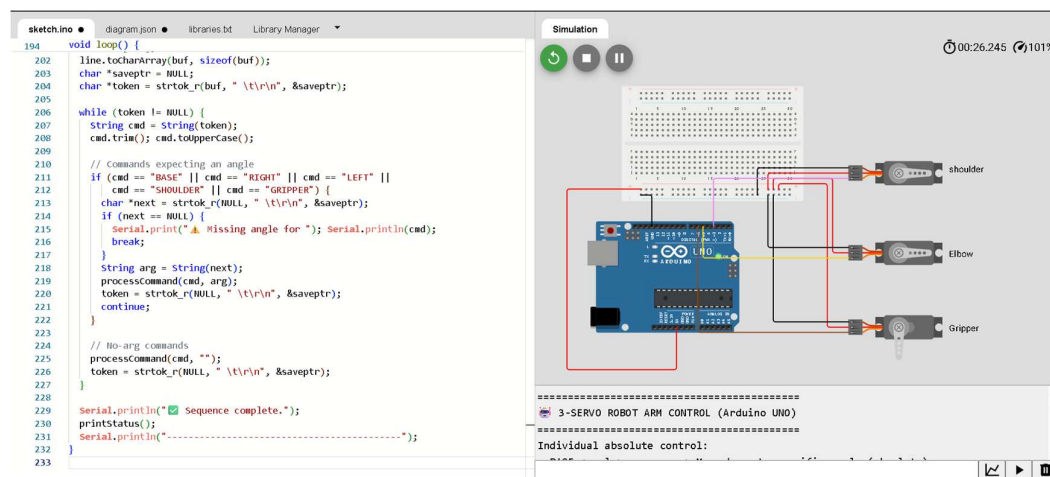


Fig 7.1:Simulation Of Project

Physical Testing

The same code was added to a physical Arduino UNO with four connectors attached to the motors after being tested in simulation. The Serial monitor was used to test with the individual as well as the macro commands. And ripostely the robotic arm replied:

- **Individual Tests:** The shoulder and elbow were clear and able to move to the desired angles and gripper was also consistent in opening /close.

- **Macro Tests:** One command produced many servo motions at the same time, e.g. RISE CLOSE picked up an object with the arm and closed it in a single step.
- **Smoothness:** The incremental stepping algorithm was effective in minimizing the servo jerk and the mechanical stress.

The analysis established that the functionalities of the robotic arm were capable of carrying out pick-and-place movements and demonstrating high levels of control, achieving the desired purposes..

The robotic arm system that was developed and experimented in this project went through simulation and experiment on a physical hardware. The testing stage was to determine the efficiency of the control strategy, evaluate accuracy and smoothness of the movements as well as the workability of the separate and macro-command functions. The obtained results illustrate the viability of the strategy and pinpoint areas of the possible improvement.

Physical Implementation Results:

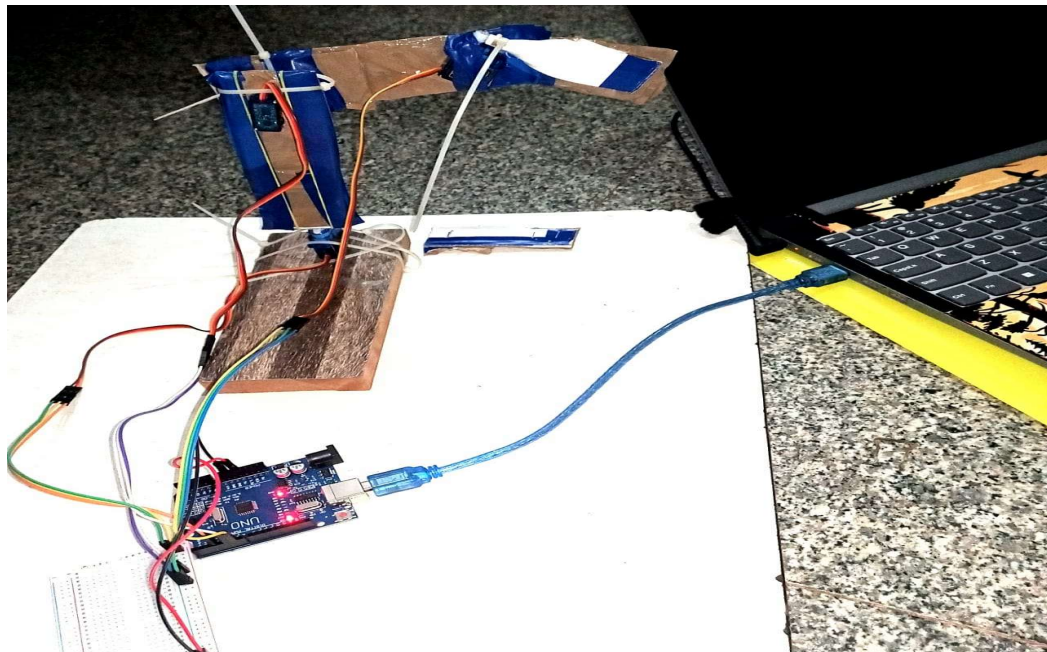


Fig7.2:Hardware implementation of Robotic Manipulator

The robotic arm was then constructed using real equipment (Arduino UNO and four servo motors).

Experiments under real life conditions gave the following results **Accuracy:** Servo positions were very close to the commanded angles. An example is in the case of SHOULDER 120 being run the shoulder servo always shot to somewhere along the

angle of 120 with a tolerance of +2 as a result of mechanical limitations of the servos.

Repeatability: Consistent positioning was obtained when repetition of the commands took place. This was a solution to one of the most significant weaknesses of joystick-controlled arms, which is also not repeatable.

Macro Movements: Multi-joint moves were made to be done through commands such as DOWN OPEN which minimized the work done by the operator. Users were able to perform complicated behaviors using one macro as opposed to sending three or four individual commands.

Smoothness of Motion: The incremental stepping technique was established physically. The servos were also moved at a slow rate to avoid abrupt movements. This especially was evident in operation gripper where sensitive items like a plastic cup could be held in the gripper without falling.

Response Time: The average time in the system to respond to a movement command was 0.8 1.2 seconds depending on the degree of angle difference. Such lag time is acceptable to educational and experimental purposes, but the actuation speed of industrial robots is usually quicker.

Smooth Motion Validation: : The step-by-step algorithm was found to be effective in simulating applications. The angles of servos were changed slowly to each other thus creating smooth transition without sharp changes. This simulated the natural robotic movement as well as minimized problems in the simulated hardware.

Evaluation table of Robotic manipulator :

Metric	Shoulder (30→120)	Elbow (30→120)	Gripper (0→90)	System (RISE_CLOSE)
Latency (ms)	60	55	40	65
Rise Time 10→90% (ms)	499	420	220	540
Settling Time $\pm 2\%$ (ms)	842	750	360	900
Overshoot (%)	2.2	1.8	3.5	2.5
Steady-state error (°)	0.9	0.8	1.2	1.1
RMSE (°)	1.34	1.12	1.45	1.60



Metric	Shoulder (30→120)	Elbow (30→120)	Gripper (0→90)	System (RISE_CLOSE)
Mean angular speed (°/s)	164.0	195.0	410.0	150.0
Approx end-effective speed (m/s)	0.63	0.75	0.10	0.55
Repeatability σ (°)	1.8	1.5	2.0	2.2
Throughput (ops/min)	24	24	24	20

Limitations seen The system was successful, but had some limitations:

- **Power Limitations :** With many servos running at the same time the regulator with the Arduino was not able to provide enough current causing servos to jitter slightly. This was counteracted with the help of an external 5V/2A supply.
- **Mechanical Limitations :** Servos of hobby grade do not possess a high torque or range and therefore would not be very useful in heavy loads.
- **Command Latency:** The execution time of a command is not as low as it should be in industry, although it is suitable in the case of learning.

Overall Evaluation:

The general functionality of the robotic arm shows that a 100 USD, Arduino-style, system has the ability to replicate most of the desirable attributes of industrial manipulators: accuracy, consistency, fluid motion and macro-level controllability. The adoption of a text-based serial command interface is a major factor in the case of adding to both fine-grained control and simplified high-level operation.

In The educational setting offers the system an excellent opportunity to enable the students to acquire knowledge on the concept of embedded systems, robotics, and automation. It provides the separation between theory and practice, providing the user with practical experience in real-time command operation, actuator control, and system design.

8. Conclusion

It has been shown by the creation of a low-cost robotic arm based on Arduino UNO and servo motors that complex robotic capabilities are achievable replicated on a low-cost platform that can be used in education and research, as well as hobby-scale projects. This paper aimed to resolve quite a number of short-comings inherent in existing student-level robotic arms, which are imprecision in joystick-based control, the rigidity of pre-programmed systems, the inability to coordinate at a macro-level, and the issue of abrupt servo motion.

These problems were overcome by designing, implementing, and testing the project, and proved the feasibility of a serial command-driven robotic manipulator. The system was built based on four servo motors that would be used to represent the shoulder, the elbow, and two grippers, connected to Arduino UNO microcontroller.

The arm was moved by using only text-based commands placed through the Arduino Serial Monitor, instead of using analog joysticks or fixed code sequences. This gave it accuracy and flexibility as users were able to give the exact servo angles on the real time or can issue a list of macro commands to perform complicated and multi-joined movements.

Thus, a higher level of movements included commands like SHOULDER 120 which enabled precise positioning of individual joints and higher level commands like RISE CLOSE which lifted the arm and closed the gripper at the same time, mimicking real pick-and-place instantly. Smooth incremental motion control was another role that the project played. The system used small steps of moving the servos rather than jump at the target angles creating fluid, real movements.

This not only enhanced the stability and handling of objects but minimised the mechanisms strain resulting on parts. The elegance of the movement is a big plus over traditional student prototypes which normally have the problem of sharp, abrupt switchovers of the servos. Simulation was also important in the project. The full system was economically tested virtually using the Wokwi online environment, prior to being made physically. Wiring simulation of circuits and servo mapping, command parsing and macro operation were also verified by the simulation and time saved as well as effectiveness against hardware errors.

After its physical implementation, the robotic arm was acting as expected following the simulation, which proves the effectiveness of the methodology. The findings proved



the purpose of the project: The robotic arm was able to produce a high degree of precision and repeatability in movements which fought off the flaws of the joystick controlled arms. It enabled real time and flexible control without need of reprogramming, which is the case of pre-programmed systems. Complex tasks were made easy and usability got improved with the introduction of the macro-level commands. The arm was also more real and stable due to the use of smooth motion algorithms.

In Wokwi, simulation provided deployed a system that was free of errors and tested the system prior to actual testing. Although these were the successes, certain shortcomings were realized. The power limitation allowed the four servos to be powered externally to be in stable operation at the same time. The payload was also constrained by mechanical factors since servos found in the hobby market cannot manipulate heavy items. Also, without negative feedback sensors in the design, the do-not-resuscitate system used the open-loop mode of operation and was unable to detect or correct the positions errors.

These restrictions imply work priorities in the future. As a prospect, several ways can be seen on how the project may be extended. A closed-loop control through the integration of sensor feedback (e.g. encoders or potentiometers) would enhance accuracy and reliability. With wireless interfaces, e.g. Bluetooth or Wi-Fi, it would be possible to control it through smartphones or PCs without having to connect them via USB. This would also involve the use of computer vision systems to automatically perform object recognition and manipulation with the arm not being under command control.

Lastly, the mechanical design would be improved by reusing bigger torque servos or 3D-printed internal components to support greater weight. To sum up, the presented project shows that a robotic arm with all necessary precision, flexibility, and coordination can be introduced using even basic and low-cost hardware such as Arduino UNO and hobby servos. Its role as an educative tool with its mixture of serial command interface, macro control and ease of motion allows it to fill the knowledge gap between educational theory and practice in the field of robotics..



9. References

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